Shuo Ma

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Research Areas

Experimental atomic, molecular, and optical physics; optical atomic clocks; neutral-atom quantum information; quantum sensing and precision measurement.

Research Appointments

Postdoctoral Researcher, University of California, Berkeley 2024–present
Advisor: Shimon Kolkowitz

- Developing a meter-scale, multiplexed ^{87}Sr optical lattice clock for synchronous differential measurements and precision tests of gravitational redshift.
- Demonstrated Aubry–André localization in a bichromatic optical lattice, improving Ramsey coherence by up to 100-fold in a horizontal clock (arXiv:2610.17499, 2026).
- Demonstrated erasure conversion to extend optical-clock coherence (*PRX Quantum*, 2025).

Graduate Researcher, Princeton University 2018–2024
Advisor: Jeff D. Thompson

- Established a ^{171}Yb optical-tweezer-array platform for neutral-atom quantum computing.
- Demonstrated nuclear-spin qubit control, high-fidelity two-qubit gates, and mid-circuit erasure conversion (*Physical Review X*, 2022; *Nature*, 2023).

Undergraduate Researcher, University of California, Santa Barbara 2016–2018
Advisor: David M. Weld

- Designed and built a narrow-linewidth optical reference cavity; proposed an atomic-crucible design for rapid atomic spectroscopy.

Education

Ph.D. in Physics, Princeton University 2024
Thesis: *Quantum Computing with Neutral Yb Atom Arrays*
Advisor: Jeff D. Thompson

B.S. in Physics, University of California, Santa Barbara 2018
Highest Honors; GPA: 3.91
Thesis: *New Experimental Approaches in Atomic Physics, Quantum Interfaces, and Energy Harvesting Technology*
Advisor: David M. Weld

Honors and Awards

2025 Finalist, APS Division of Quantum Information Best Dissertation Award
2018 Academic Excellence Award, UC Santa Barbara

Publications and Preprints

My name appears in bold. An asterisk denotes equal contribution.

1. **S. Ma***, A. Stöltzel*, B. Christensen, Z. Ding, D. Ganapathy, A. Householder, G. Liu, and S. Kolkowitz, "An Aubry–André localized optical lattice clock," preprint, arXiv:2610.17499 (2026).
2. B. Zhang, G. Liu, G. Bornet, S. P. Horvath, P. Peng, **S. Ma**, S. Huang, S. Puri, and J. D. Thompson, "Logical qubits with erasure conversion using metastable neutral atoms," *Nature Physics* **22**, 910–916 (2026).
3. **S. Ma***, J. Dolde*, X. Zheng*, D. Ganapathy, A. Shtov, J. Chen, A. Stöltzel, B. J. Christensen, and S. Kolkowitz, "Enhancing optical lattice clock coherence times with erasure conversion," *PRX Quantum* **6**, 040340 (2025).
4. J. Dolde, D. Ganapathy, X. Zheng, **S. Ma**, K. Beloy, and S. Kolkowitz, "Direct measurement of the $^3P_0$ clock-state natural lifetime in ^{87}Sr," *Physical Review A* **112**, 023121 (2025).

5. M. Peper*, Y. Li*, D. Y. Knapp, M. Bileska, **S. Ma**, G. Liu, P. Peng, B. Zhang, S. P. Horvath, A. P. Burgers, and J. D. Thompson, "Spectroscopy and modeling of ^{171}Yb Rydberg states for high-fidelity two-qubit gates," *Physical Review X* **15**, 011009 (2025).
6. **S. Ma***, G. Liu*, P. Peng*, B. Zhang, S. Jandura, J. Claes, A. P. Burgers, G. Pupillo, S. Puri, and J. D. Thompson, "High-fidelity gates and mid-circuit erasure conversion in an atomic qubit," *Nature* **622**, 279–284 (2023).
7. **S. Ma***, A. P. Burgers*, G. Liu, J. Wilson, B. Zhang, and J. D. Thompson, "Universal gate operations on nuclear spin qubits in an optical tweezer array of ^{171}Yb atoms," *Physical Review X* **12**, 021028 (2022).
8. A. P. Burgers, **S. Ma**, S. Saskin, J. Wilson, M. A. Alarcón, C. H. Greene, and J. D. Thompson, "Controlling Rydberg excitations using ion-core transitions in alkaline-earth atom-tweezer arrays," *PRX Quantum* **3**, 020326 (2022).
9. J. T. Wilson*, S. Saskin*, Y. Meng, **S. Ma**, R. Dilip, A. P. Burgers, and J. D. Thompson, "Trapping alkaline earth Rydberg atoms in optical tweezer arrays," *Physical Review Letters* **128**, 033201 (2022).
10. C. Conner, T. de Visser, J. Loessberg, S. Sherman, A. Smith, **S. Ma**, M. T. Napoli, S. Pennathur, and D. Weld, "Energy harvesting with a liquid-metal microfluidic influence machine," *Physical Review Applied* **9**, 044008 (2018).

Invited Talks

Quantum computing with ^{171}Yb arrays 2025
APS Division of Quantum Information Best Dissertation Award Session, Los Angeles, California.

Contributed Talks

Universal quantum gates using nuclear spin qubits in an optical tweezer array of ^{171}Yb atoms 2022
APS Division of Atomic, Molecular and Optical Physics, Orlando, Florida.

Ytterbium Rydberg atom arrays 2021
APS Division of Atomic, Molecular and Optical Physics, virtual meeting.

Teaching and Mentoring

Research mentoring, University of California, Berkeley 2024–present
Mentor undergraduate and graduate researchers on optical lattice clocks, including atom-transport modeling, laser systems, and control electronics. Guided students in theoretical modeling and worked with them to complete measurements for the Aubry–André optical lattice clock project.

Research mentoring, Princeton University 2018–2024
Mentored graduate and undergraduate researchers in the neutral-atom laboratory.

Teaching Assistant, Princeton University
Quantum Mechanics with Applications (ELE 511).

Guest Lecturer, University of California, Berkeley
Graduate-course lecture on advances in neutral-atom quantum computing.

Professional Service

Referee for *Physical Review X*, *Physical Review Letters*, *PRX Quantum*, *Physical Review A*, and *Physical Review Applied*.